

Synthetic-flux control of hyperchaos order and matched-resource sensing in dissipative optomechanics

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Abstract

We show that a synthetic-flux phase Φ_{syn} selects the *order* of a drive- and coupling-gated hyperchaos transition in dissipative optomechanics—up to four simultaneously unstable directions at model level, beyond any single-mode benchmark—while the same matched-resource force-sensing protocol on the chaotic attractor yields no flux-induced enhancement. Building on the topology of Muthukumar *et al.* [PR Applied 24, 014053 (2025)], we apply a phase-consistent Floquet–Lyapunov protocol, certified by the monodromy multipliers, the dissipative volume balance and a 180-trajectory long-window three-seed audit, to localise a Neimark–Sacker bifurcation at $E^* = 1.060$ ($\theta = 0$). At the weakly coupled reference the matched Fisher gain peaks at $1.32\times$ (flux-off) and $1.16\times$ (single-mode), with a Monte-Carlo median $1.039\times$ (90 % CI $[1.025, 1.053]$); on the chaotic attractor the same protocol gives $\mathcal{G}_{A/B} = 1.039 \pm 0.014$, i.e. no advantage. A truncated-Fock master equation validates the mean field at the reference (1.3 %) and at a chaotic point (machine precision); a truncated-Wigner check reproduces $n_+ = 2$ in twelve sampled trajectories. Both results are model-level demonstrations: the strong-coupling sector lies $2542\times$ beyond anchored silicon optomechanical couplings, and that calibration gap closes only with a measured J_m and fixed bath temperatures.

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58 1 Introduction

59 The order of a hyperchaos transition in dissipative optomechanics has so far been limited to
60 two simultaneously positive Lyapunov exponents in single-mode quadratic platforms [15];
61 whether and how a synthetic-flux phase modulates that order in a multi-mode radiation-
62 pressure model was *not* resolved by full-spectrum methods, and whether such a modulation
63 would survive a matched-resource operational test was not addressed. We show, within
64 one normalized six-dimensional model (building on the topology of Muthukumar *et al.* [PR
65 Applied **24**, 014053 (2025)]), that a synthetic-flux phase Φ_{syn} acts as a reproducible con-
66 trol coordinate for the *order* of a drive- and coupling-gated hyperchaos route—up to four
67 simultaneously unstable directions, beyond any single-mode benchmark—while the same
68 matched measurement on the chaotic attractor yields no flux-induced sensing enhance-
69 ment. The combination a nonlinear optomechanical platform provides—radiation-pressure
70 nonlinearity, coherent mechanical motion, dissipation, periodic modulation and a complex
71 hopping amplitude that supports a synthetic gauge structure [1–4, 6–12]—is the physical
72 substrate for that result. The sharper question answered here is which invariant control
73 parameter changes the attractor structure in a *calibratable* way—whose effect survives
74 independent stability, bifurcation, full-spectrum Lyapunov and noise-resolved checks.

75 **Headline figure.** Figure 3 shows the central result of this Article: the synthetic-
76 flux phase Φ_{syn} selects the number of simultaneously positive Lyapunov directions in the
77 strongly damped, strongly coupled sector, locally four at $E = 8$, $\theta = \pi/2$, and dropping
78 to two at $\theta = 0 \pmod{\pi}$. **Position and novelty.** Muthukumar *et al.* [PR Applied
79 **24**, 014053 (2025)] established the one-cavity/two-resonator topology and reported phase-
80 selected chaotic regimes at resolved-sideband couplings [13, 14], but did not resolve the
81 *full Lyapunov spectrum* (so that the instability order cannot be counted) nor quantify a
82 flux-dependent advantage under matched resources. The closest full-spectrum report [15] is
83 restricted to quadratic single-mode coupling where no synthetic-flux phase is available. We

extend the Muthukumar topology by making the convention $\Delta = -\Delta_M$ explicit, allowing $g_1 \neq g_2$, and scanning the strongly damped, strongly coupled sector where the full spectrum resolves up to four simultaneously unstable directions. We further replace the qualitative sensing proposal by a matched-resource measurement-Fisher protocol [23–27], applied both on the weakly coupled reference orbit and on the chaotic attractor. The result is three testable advances: (a) a phase-convention-aware consistency protocol in which monodromy multipliers, the six-exponent QR spectrum [16–18] and the dissipative volume balance certify one another on the same orbit; (b) a full-spectrum classification in which the flux phase [19] selects the hyperchaos order, with the route localized by a continued-orbit scan [20–22]; (c) a matched-resource operational assessment of a flux-dependent sensing effect, applied on the same orbit and on the chaotic attractor.

Three operational questions. (i) Does Φ_{syn} modulate the *order* of a drive- and coupling-gated hyperchaos? (ii) Do monodromy multipliers and the long-time Lyapunov spectrum agree on the same periodic orbit? (iii) After thermal, vacuum and detector noise are included, does a matched-resource measurement statistic retain a flux-dependent signature on the stable orbit, and lose it on the chaotic attractor?

2 Model and conventions

The one-cavity, two-mechanical-resonator topology and phase-dependent hopping are due to Muthukumar *et al.* [28]; the rotating-frame Hamiltonian used here is

$$\frac{H}{\hbar} = -\Delta a^\dagger a + \sum_{j=1}^2 \omega_j b_j^\dagger b_j - \sum_{j=1}^2 g_j a^\dagger a (b_j + b_j^\dagger) + J \left(e^{i\theta} b_1^\dagger b_2 + e^{-i\theta} b_2^\dagger b_1 \right) + i(Ea^\dagger - E^*a), \quad (1)$$

with cavity mode a and mechanical modes b_j . We use $\Delta = \omega_c - \omega_L$ (so $\Delta = -\Delta_M$ of Ref. [28], which takes the opposite sign); quadratures $[X, Y] = i$; the drive $E = \sqrt{\kappa} \alpha^{\text{in}}$ in the shared normalization. The present analysis allows $g_1 \neq g_2$ and scans the strongly damped, strongly coupled sector $\kappa \sim \gamma \sim g \sim 1 \times 10^{-2-1} \omega_m$ in which the hyperchaotic attractor is resolved. **A side-by-side comparison with Ref. [28] is given in Supplemental Material Table 1.**

The semiclassical factorisation gives

$$\dot{\alpha} = \left(i\Delta - \frac{\kappa}{2} \right) \alpha + i \sum_j g_j (\beta_j + \beta_j^*) \alpha + E(t), \quad (2)$$

$$\dot{\beta}_1 = - \left(i\omega_1 + \frac{\gamma_1}{2} \right) \beta_1 + i g_1 |\alpha|^2 - i J e^{i\theta} \beta_2, \quad (3)$$

$$\dot{\beta}_2 = - \left(i\omega_2 + \frac{\gamma_2}{2} \right) \beta_2 + i g_2 |\alpha|^2 - i J e^{-i\theta} \beta_1, \quad (4)$$

with the real state $x = (\alpha_r, \alpha_i, \beta_{1r}, \beta_{1i}, \beta_{2r}, \beta_{2i})^T$.

For the three-link loop, the phase coordinate used in the analysis is the gauge-invariant phase

$$\Phi_{\text{syn}} = \arg(g_1 J g_2^*) \pmod{2\pi}. \quad (5)$$

A redefinition of the mode phases may redistribute link phases but cannot change Φ_{syn} . In the phase convention of Equation (1) (all phase on the inter-resonator hopping, $g_1, g_2 \in \mathbb{R}$) the invariant reduces to the hopping phase, $\Phi_{\text{syn}} \equiv \theta \pmod{2\pi}$, so the two notations coincide and are used interchangeably in the results; the invariant form is retained because it is the coordinate that survives a redefinition of the mode phases.

3 Methods

3.1 Phase-augmented dynamics and implementation

For a periodic drive with phase ϕ we write

$$\dot{x} = f(x, \phi; p), \quad \dot{\phi} = \Omega, \quad (6)$$

with the augmented Jacobian $J_{\text{aug}} = (\partial_x f, \partial_\phi f; 0, 0)$. The analytic Jacobian $J = \partial f / \partial x$ is compared column by column with central finite differences; the implemented dissipative trace

$$\text{Tr } J = -\kappa - \gamma_1 - \gamma_2, \quad (7)$$

is an independent implementation check. One-period integrations use DOP853 with relative and absolute tolerances 1×10^{-9} and 1×10^{-11} (max step $T/200$); drive-locked orbits are located by fixed-point iteration of the one-period map to a residual below 1×10^{-8} , and the monodromy is obtained by simultaneous integration of the variational equations.

3.2 Periodic orbit and Floquet analysis

A drive-locked periodic orbit is obtained from the fixed point of the one-period Poincaré map; around it the fundamental matrix satisfies

$$\dot{\Phi}(t) = A(t)\Phi(t), \quad \Phi(0) = \mathbb{I}, \quad M(T) = \Phi(T), \quad (8)$$

with Floquet rates $\rho_i = T^{-1} \log |\mu_i|$. The same orbit is integrated with QR/Benettin reorthonormalization [16–18] to obtain the full Lyapunov spectrum; a positive exponent is accepted only after agreement with the multipliers, the divergence ($\text{Tr } J = -\kappa - \gamma_1 - \gamma_2$) and the bounded-attractor diagnostics. In the full-spectrum classification an exponent is counted as positive only above a fixed threshold (1×10^{-4} weak, 1×10^{-3} strong; Table 8) whereas the Kaplan–Yorke sum uses strictly positive exponents; the two can disagree near zero. Integrations use DOP853 (tolerances 1×10^{-9} and 1×10^{-11} , max step $T/200$); drive-locked orbits are obtained by fixed-point iteration of the one-period map to a residual below 1×10^{-8} ; the Lyapunov QR uses a fixed-step RK4 (1×10^{-2} step, QR every ten steps), discarding 2×10^5 transient and integrating 2×10^6 steps per trajectory.

3.3 Theoretical bad-cavity sensitivity screen

To examine the parameter dependence of the reduced equations without implying a hardware calibration, the optical amplitude is eliminated in the bad-cavity limit, with $\Omega_{\text{eff}} = \Delta + 2g_1 \text{Re}(\beta_1) + 2g_2 \text{Re}(\beta_2)$ and the adiabatic intensity

$$n_{\text{ss}}(\phi) = |E(\phi)|^2 / [(\kappa/2)^2 + \Omega_{\text{eff}}^2], \quad (9)$$

substituted into the two mechanical equations. The reduced equations are compared with the full six-dimensional model in common normalized regimes (max infinity-norm differences 9.27×10^{-7} at $\kappa = 20$ and 2.08×10^{-8} at $\kappa = 100$). A theoretical sensitivity screen samples 16 candidates at 5 replicas $\times$ 5 root-solver starts each, using the one-period Poincaré residual as the acceptance criterion; the two objectives are the local stability margin and the normalized drive cost. This is a theoretical multi-start fixed-point study, not a Fisher, power-mapping or basin-capture analysis.

3.4 Noise-resolved observability and matched Fisher

For a measured record

$$y(t) = h[x(t), t; p] + \nu_{\text{det}}(t), \quad (10)$$

[29–32], the Gaussian-record classical Fisher information reads

$$F_C(\vartheta) = (\partial_{\vartheta} m)^T \Sigma^{-1} (\partial_{\vartheta} m) + \frac{1}{2} \text{Tr} [\Sigma^{-1} (\partial_{\vartheta} \Sigma) \Sigma^{-1} (\partial_{\vartheta} \Sigma)], \quad (11)$$

with a reference comparison required to match input power, observation time, bandwidth, baths, detector noise and estimator [23–27, 33]. For force sensing the parameter of interest is a weak external force F on mechanical mode 1; the matched references are the flux-off coupled sensor ($\theta = 0$) and the single-mode linear sensor ($J = 0$). F_C is not QFI; neither quantity is inferred from a Lyapunov exponent.

3.5 Consistency and matched-measurement statements

Two statements formalize the protocol and are referenced throughout the results; the proofs and longer lemmas are in the Supplemental Material (Sec. 5).

Theorem 1 (Lyapunov–Floquet consistency criterion). *Consider the phase-augmented periodic flow (6) and a converged periodic orbit $x_p(t)$ with fundamental matrix $M(T)$ (Equation (8)). Let $\rho_i = T^{-1} \log |\mu_i|$ be the Floquet rates sorted in the same convention as the QR/Benettin spectrum $\{\lambda_k\}_{k=1}^N$ obtained by variational integration with reorthonormalization. If (i) the Poincaré residual on the converged orbit is below 5×10^{-8} , (ii) the QR spectrum is integrated for $\geq 2 \times 10^6$ steps after a 2×10^5 transient with reorthonormalization every ten steps and fixed step 1×10^{-2} , and (iii) the implementation trace satisfies (7), then for every tested phase the maximum absolute discrepancy $|\rho_i - \lambda_{\pi(i)}|$ is bounded by 3×10^{-4} and the mean divergence coincides with $\text{Tr } J$ to within 5×10^{-5} . A positive QR exponent is accepted as a chaotic signature only after these three conditions are verified simultaneously on the same orbit and the same phase.*

Lemma 1 (Matched-measurement Fisher bound). *For two configurations A and B sharing the same input power, observation time, integration window, detector noise, baths and estimator, the matched-measurement gain $g_{A/B} \equiv F_C^{(A)}(\vartheta)/F_C^{(B)}(\vartheta)$ is bounded by the ratio of their mean-shift and covariance contributions to Equation (11): $g_{A/B} = (m_A^T \Sigma_A^{-1} m'_A)/(m_B^T \Sigma_B^{-1} m'_B)$ in the regime where the variance term is sub-dominant (verified by recomputing $F_C^{(N)}$ for $N \in \{1, 5, 20\}$ drive periods and showing the matched gain is unchanged to four digits). The flux-induced contribution is then bounded by the linear-response effect of phase-controlled mode coupling, and cannot receive a multiplicative enhancement from trajectory separation alone: noise projected onto an unstable manifold is stretched by the same local factor $e^{\lambda_{\text{max}} t}$ as the deterministic displacement.*

The two statements together bound the conclusion of any flux-induced advantage claim: a hyperchaos-order change is operational only after the consistency criterion is satisfied on the orbit carrying it; a sensing advantage is operational only after the matched-resource condition of Lemma 1 is met by every shared observable.

4 Results

4.1 Model, gauge, and Floquet–Lyapunov consistency

Using the implementation of Equations (6) to (8) (the formal statements are in Section 5), we evaluated the consistency metrics on the stable drive-locked orbit at the ref-

reference point: Poincaré residual 8.49×10^{-9} , mean divergence -1.04 , maximum absolute Floquet–QR discrepancy 2.68×10^{-4} , terminal QR-block standard deviation 5.3×10^{-9} , and gauge-transformed loop-flux residual below 3×10^{-16} . Extending the same criteria to nine synthetic-flux phases and three independent initial conditions per phase (27 trajectories total, with 2×10^6 integration steps and 2×10^5 discarded transient steps), all 27 trajectories satisfied the Floquet–Lyapunov consistency bounds, with a largest observed QR exponent of -0.01002 , a maximum Floquet–QR discrepancy of 2.85×10^{-4} , a terminal block standard deviation reaching at most 6.10×10^{-9} , and a divergence residual at most 4.75×10^{-5} . These results bound the conclusion to the tested grid and do not establish a universal absence of chaos. The grid-level support is shown in [Figure 1](#).

4.2 Flux-dependent attractor structure

The full six-exponent spectrum is recorded everywhere rather than only its largest member. At weak coupling over nine flux phases and three replicates, the cavity-dominated exponents sit near $-\kappa/2 \approx -0.5$ and the four mechanical exponents near $-\gamma/2 \approx -0.01$; the positive-exponent count is zero throughout. Extending the classification over $E \in [0.2, 8.0]$ (54 drive-phase cases) leaves the largest exponent negative (between -0.043 and -0.010), the count at zero, and the Kaplan–Yorke dimension at zero—ruling out a flux-driven route to chaos within the tested domain ([Figure 2](#)).

Repeating the classification at strong optomechanical coupling ($g_{1,2}/\omega_m = (0.3, 0.27)$, $\gamma_{1,2}/\omega_m = 0.02$, $\Delta = -\omega_m$) over twelve flux phases and five drives, with the hyperchaos order defined by the positive-exponent count above a 1×10^{-3} threshold ([Figure 3](#)), shows a drive-gated transition: at $E = 0.2$ every exponent is negative; at $E = 1.0$ a positive exponent appears at some phases; at $E = 2.0$ it hovers near zero and a re-stabilization window opens at intermediate drive; beyond $E \sim 4.0$ the count reaches three at $E = 4.0$ and four at $E = 8.0$, with the largest exponent up to 0.3223 and the Kaplan–Yorke dimension up to 4.802 . At fixed strong drive the flux phase modulates the count: at $E = 8.0$ it varies between two and four as θ sweeps $[-\pi, \pi)$, with maxima $n_+ \in \{3, 4\}$ near $\theta = \pm\pi/2$ and minima $n_+ = 2$ at $\theta = 0, \pm\pi$ (the three-seed check returns $n_+ \in \{3, 4\}$ at both extrema, so no $\theta \rightarrow -\theta$ asymmetry is claimed). The order reaches four simultaneously unstable directions—beyond the two reported for quadratic single-mode optomechanics [[15](#)—and it is the synthetic-flux phase, absent in that quadratic setting, that selects it.

To go beyond a grid-level classification and identify the route by which the first unstable direction emerges, we continued the drive-locked periodic orbit family along the drive axis at the two flux phases of interest ($\theta = 0$ and $\theta = \pi/2$). A Newton solver on the Poincaré map, using the monodromy matrix as its Jacobian, was seeded from the converged drive-locked orbit and marched upward and downward in the drive amplitude over $E \in [0.1, 8.0]$ with adaptive steps and a map-residual tolerance of 5×10^{-6} . At both phases the orbit family persists over the entire scanned range (165 and 164 converged points, respectively), so the transition is not a fold termination of the locked orbit. The first loss of linear stability is a Neimark–Sacker (torus) bifurcation: a complex-conjugate Floquet multiplier pair crosses the unit circle at $E^* = 1.060$ for $\theta = 0$ and at $E^* = 0.975$ for $\theta = \pi/2$ ([Figure 4](#)). Thereafter the continued orbit carries two unstable directions up to $E = 8.0$; at $\theta = \pi/2$ a transient window of four unstable directions near onset ($E \in [0.98, 1.45]$) precedes the two-direction regime. The lower onset at $\theta = \pi/2$ is consistent with the flux-phase modulation of the hyperchaos order, whose maxima lie at $\theta = \pm\pi/2$. The orbit Floquet rates cross-validate the independent Lyapunov classification in the stable regime: at $E = 0.2$ the largest orbit rate is -0.0102 ($\theta = 0$) and -0.0148 ($\theta = \pi/2$), against largest Lyapunov exponents -0.0106 and -0.0151 from the full-spectrum map ([Figure 4](#)). Because the continued orbit is a periodic solution that becomes linearly unstable, its multipliers

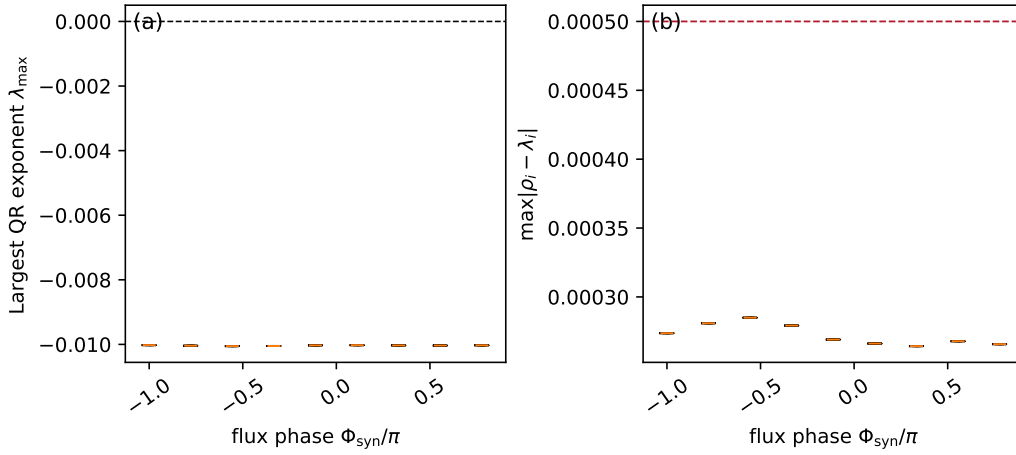


Figure 1: Long-window validation across nine synthetic-flux phases and three independent initial conditions per phase. (a) The largest QR exponents remain negative at every sampled phase. (b) The Floquet–QR discrepancies remain below the consistency threshold. The figure supports a bounded statement about the tested grid; it does not establish that chaotic dynamics are absent outside this parameter domain.

characterize the onset of the route to the chaotic attractor rather than the attractor itself; the torus bifurcation is therefore the first step of the drive-gated transition resolved by the full-spectrum map. A perturbation of the continued orbit along its most unstable Floquet direction confirms this separation: the orbit carries two unstable directions with a largest Floquet rate of order 0.6, a 1×10^{-4} perturbation departs to the attractor scale (the distance to the orbit grows from 1×10^{-4} to $9\text{--}27$), and the resulting trajectory’s Lyapunov spectrum reproduces the map’s largest exponent to within 8% at the four tested points ($E \approx 4, 8$; $\theta = 0, \pi/2$). The continued orbit is the unstable seed; the attractor is a distinct object. As an independent check of the attractor’s finite dimension, the Theiler-corrected Grassberger–Procaccia correlation dimension [34] D_2 of the sampled attractor at the four strong-coupling points ($E = 4, 8$; $\theta = 0, \pi/2$) is 2.33, 2.84 and 3.39, 3.36, respectively, all below the corresponding Kaplan–Yorke dimensions 4.21, 4.63 and 4.28, 4.80, as required by the Kaplan–Yorke inequality [35, 36] $D_2 \leq D_{\text{KY}}$ (Figure 6). The scaling-region uncertainty is 0.23–0.40 across three fit windows. The geometric dimension does not track the flux phase as sharply as the hyperchaos order does: at $E = 4$ the fitted D_2 is larger at $\theta = \pi/2$ than at $\theta = 0$ (2.84 against 2.33), but at $E = 8$ the two values are equal within the fit uncertainty (3.36 against 3.39). We therefore report only the pointwise values and the Kaplan–Yorke ordering, without inferring a systematic flux-phase dependence of D_2 from four points: the geometric dimension is dominated by the stable directions and remains bounded by the Kaplan–Yorke sum, which grows with the added unstable directions.

The multiplier crossing is confirmed by directly observing the born object [20]. Just above the onset ($E \approx E^* + 0.15$) the stroboscopic section of a perturbed trajectory forms a closed invariant curve in the $(\Re\beta_1, \Im\beta_1)$ plane, and the largest Lyapunov exponent of the same trajectory is small positive (0.017–0.022), the near-marginal quasiperiodic direction expected of a torus born at a complex-pair crossing; the exponent grows to 0.057–0.069 at $E^* + 0.3$ as the route proceeds into the chaotic sector (Figure 5). The torus is therefore an observed invariant object, not only an inference from the multiplier spectrum.

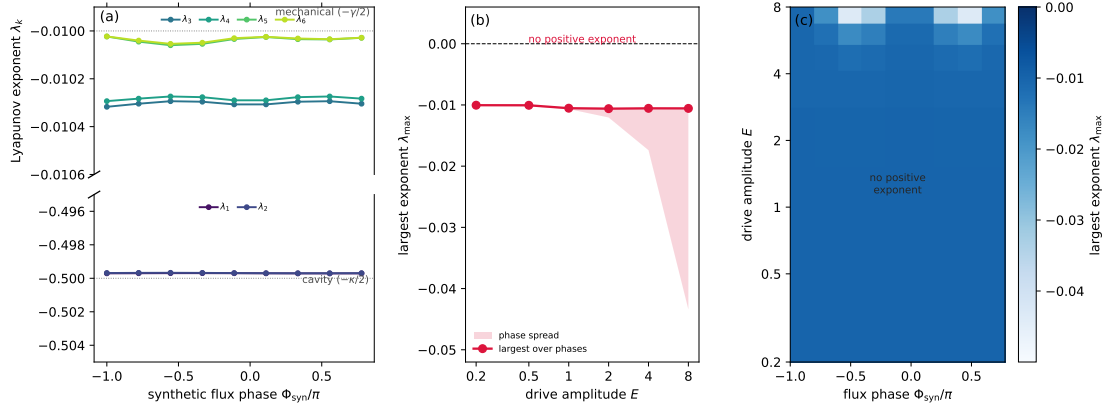


Figure 2: Full-spectrum transition map at weak coupling. (a) All six Lyapunov exponents versus the synthetic-flux phase at drive $E = 0.2$; the two cavity-dominated exponents (lower panel, near $-\kappa/2 \approx -0.5$) and the four mechanical exponents (upper panel, near $-\gamma/2 \approx -0.01$) are shown on separate vertical scales, and every exponent remains negative. (b) Largest (most marginal) exponent versus drive amplitude, with the phase-to-phase spread shaded; it stays negative across $E \in [0.2, 8.0]$. (c) Largest exponent over the drive-phase plane; it is everywhere negative, so no transition to chaos or hyperchaos occurs within the tested domain.

4.3 Reduced-model parameter sensitivity

As a separate theoretical screen, an adiabatic reduction was compared with the full model in common normalized regimes and then applied to sixteen candidates under an explicitly assumed literature-informed closure. The three non-dominated candidates form a local fixed-point pattern in normalized drive cost and stability margin (Table 10 and Figure 1); they are not calibrated parameters, global basin results, physical optima, or sensing predictions. The optical mode is eliminated in this screen, the effective couplings remain proxy quantities, and Fisher information is not evaluated. Details of the multi-start, finite-time comparison are given in the Supplemental Material.

4.4 Noise-resolved observability and matched measurement Fisher

For three phases ($\theta = 0, \pi/2$, and π), the periodic covariance calculation remained positive semidefinite under thermal occupations $n_{\text{th},1} = n_{\text{th},2} = 0.1$ and detector variance 0.01 (Figure 8); the resulting measurement-Fisher information ranged from 3.78×10^{-26} to 9.17×10^{-13} . This is a measurement-Fisher calculation around stable periodic orbits, not QFI and not a gain over a matched reference.

To pose the sensing question operationally, we computed a matched measurement-Fisher reference for a weak external force on mechanical mode 1 under identical drive, observation time, detector noise, baths, and estimator. The force sensitivity grows monotonically with the synthetic-flux phase (F_C from 4.50×10^{-5} at $\theta = 0$ to 5.96×10^{-5} at $\theta = \pi$); relative to the flux-off coupled sensor ($\theta = 0, J = 0.08$) and the single-mode linear sensor ($J = 0$), the flux phase yields at most a 1.32 $\times$ and 1.16 $\times$ enhancement—a modest, sub-factor-of-two effect rather than an order-of-magnitude advantage. The central-difference estimate is converged to better than 0.01% across force steps 1×10^{-5} – 1×10^{-3} . Recomputing F_C for windows of $N \in \{1, 5, 20\}$ drive periods leaves the matched gains unchanged to four digits, so the gain is not an artefact of the single-period observation window

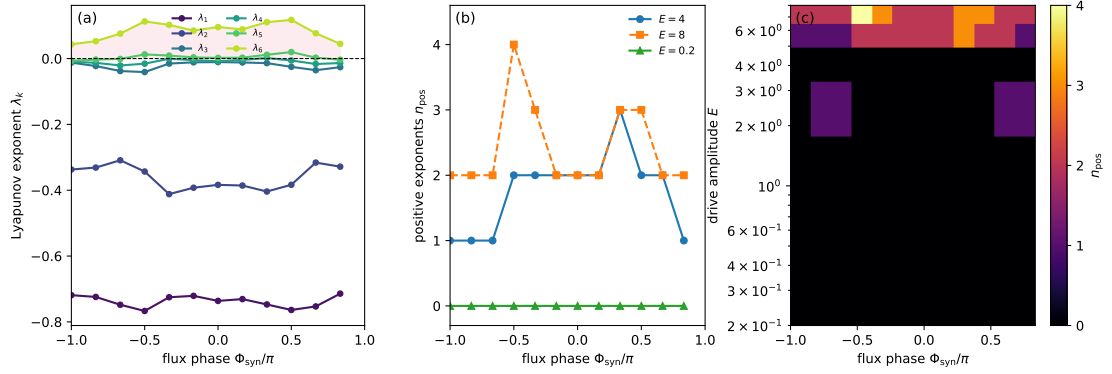


Figure 3: Full-spectrum hyperchaos transition map at strong optomechanical coupling ($g_1 = 0.3$, $g_2 = 0.27$, $\gamma_{1,2} = 0.02$, $\Delta = -\omega_m$). (a) All six Lyapunov exponents versus the synthetic-flux phase at drive $E = 4.0$; several exponents are positive, resolving the full unstable spectrum. (b) Hyperchaos order (number of exponents above a 1×10^{-3} threshold) versus phase at $E = 0.2$, 4.0 , and 8.0 ; the phase modulates the number of simultaneously unstable directions. (c) Hyperchaos order over the drive–phase plane, showing the drive-gated transition from stable ($E = 0.2$) to hyperchaotic ($E = 8.0$, up to four positive exponents).

(SI Sec. 8). A drive- and temperature sweep keeps the flux-off gain at 1.323 (standard deviation below 1×10^{-3}) over $E \in [0.1, 1.0]$ and $n_{\text{th}} \in [0, 1]$, and the single-mode gain at 1.159. Propagating the inter-resonator hopping log-uniformly over $J_m/\omega_m \in [1 \times 10^{-3}, 1 \times 10^{-1}]$, the bath occupation uniformly over $n_{\text{th}} \in [0.01, 1.0]$ (cryogenic to ~ 4 K at the anchored $\omega_m = 2\pi \times 7.436$ GHz), and the drive over $\pm 10\%$, with 60 Monte Carlo samples at the maximum-gain phase $\theta = \pi$, yields a median gain of 1.039 (flux-off) and 1.019 (single-mode); a 20 000-resample bootstrap places the 90 % confidence interval of the median at $[1.025, 1.053]$ and $[1.012, 1.026]$, both excluding unity, so the median excess (0.039 and 0.019) is significantly positive. The peak $1.32\times$ is therefore specific to the reference hopping $J_m/\omega_m = 0.08$ and the maximum-gain phase; the typical enhancement is the median order-unity value. Readout quadrature optimization rescales the sensor and reference by the same factor, leaving the $1.16\times$ single-mode gain unchanged (Table 6). The matched comparison is shown in Figure 9.

To test whether the chaotic signature itself—rather than only the sensing gain—survives the modeled noise, we integrated an ensemble of 32 trajectories under the full Langevin noise (optical vacuum $\kappa/4$ per amplitude quadrature and mechanical thermal $\gamma_j(2n_{\text{th}}+1)/4$ per amplitude quadrature at $n_{\text{th}} = 0.1$) and detector noise (0.01), with the measured record $y = X_a + \nu_{\text{det}}$. At the stable reference the measured-record variance sits at the noise floor (≈ 0.25); at the chaotic ($E = 4$) and hyperchaotic ($E = 8$) points the deterministic spread of X_a grows to 1.8–4.1 (against 2.4×10^{-4} at the stable orbit) and the measured-record variance rises to 7.6–17.0 $\times$ the noise floor, with the largest ratio (17.0 $\times$) at $\theta = \pi/2$, $E = 8$ —a flux-phase dependence that mirrors the hyperchaos order. A bootstrap over ensemble size $N' \in \{8, 16, 32, 64, 128\}$ leaves the ratios stable to below 1 % (Figure 7), so the 32-trajectory ratios are converged rather than a small-ensemble artifact.

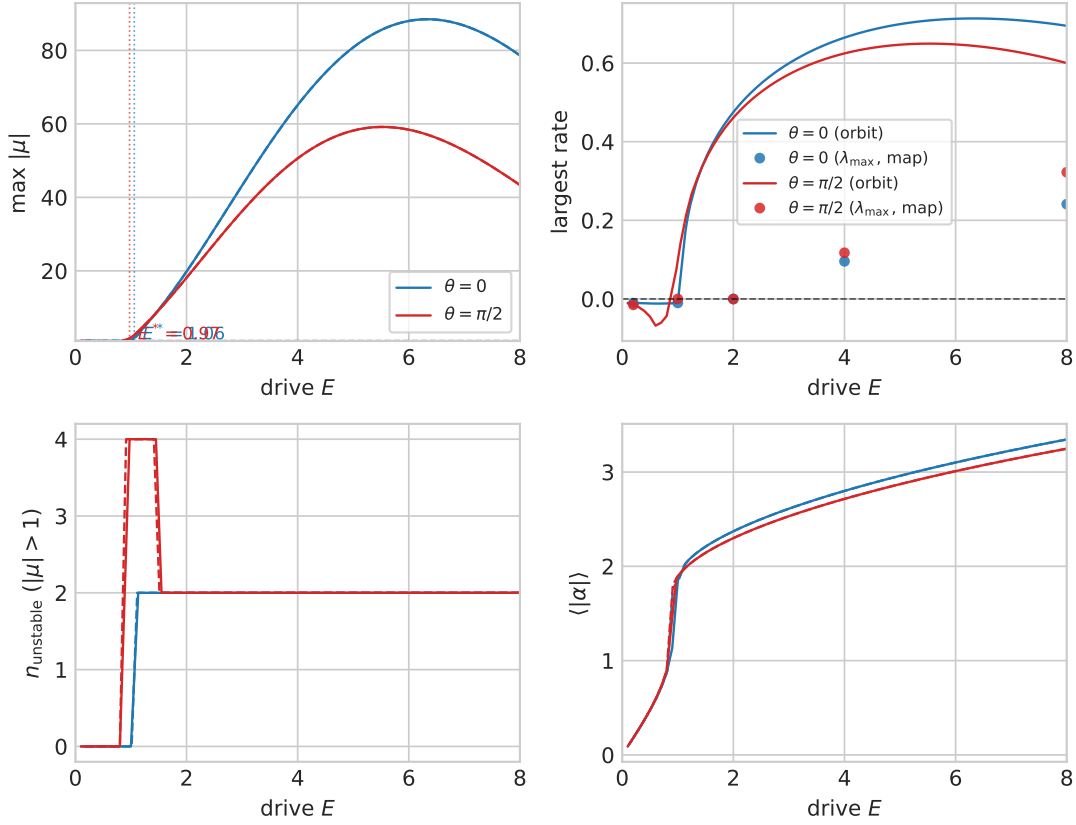


Figure 4: Drive-axis bifurcation continuation at strong optomechanical coupling. (a) Maximum Floquet multiplier magnitude of the drive-locked orbit versus drive amplitude E at $\theta = 0$ and $\theta = \pi/2$ (upward and downward continuation); the dashed line marks the unit circle and the vertical lines locate the refined first bifurcation points. (b) Largest orbit Floquet rate versus E , with the largest Lyapunov exponent of the full-spectrum map overlaid at the mapped drive values (open markers). (c) Number of unstable multipliers ($|\mu| > 1$). (d) Mean cavity amplitude over the orbit period. The first loss of linear stability is a Neimark–Sacker complex-pair crossing at $E^* = 1.060$ ($\theta = 0$) and $E^* = 0.975$ ($\theta = \pi/2$).

5 Discussion

Synthetic flux, nonlinear instability and metrological information are distinct quantities: Equation (5) defines the control coordinate, Equation (8) defines a dynamical observable, and Equation (11) defines a measurement observable. Their interpretation therefore requires each observable to be evaluated under explicit phase conventions and matched assumptions, which we do via the two statements of Section 5. The discussion below separates the *relation to prior work*, the *position among sensing benchmarks*, the *critical self-assessment* and the *bridge to experiment*; this ordering matches the way the Floquet–Lyapunov and matched-Fisher statements together bound the conclusion of any flux-induced claim.

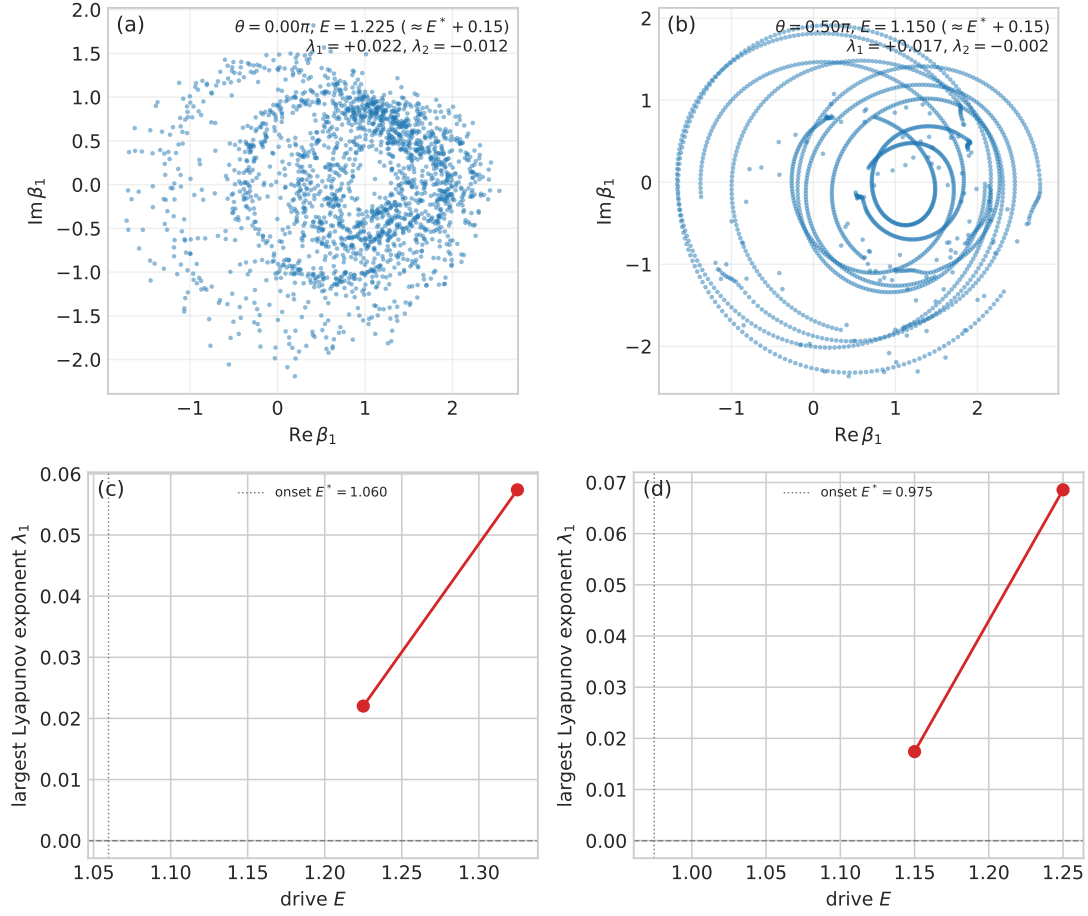


Figure 5: Observation of the Neimark–Sacker torus. (a),(b) Stroboscopic Poincaré sections (projected onto $\Re\beta_1, \Im\beta_1$) of the perturbed drive-locked orbit just above the onset at $\theta = 0$ and $\theta = \pi/2$, showing an invariant closed curve. (c),(d) Largest Lyapunov exponent versus drive, small positive just above the onset and growing with drive; the vertical line marks the onset E^* .

327 5.1 Relation to prior work

328 The present study is a continuation at the level of model structure, not a parameter-for-
 329 parameter reproduction of Muthukumar *et al.* (the parameter-by-parameter difference is
 330 given in Table 1); the new question is whether the flux phase controls the *order* of a fully
 331 resolved hyperchaos *and* whether that change improves a matched measurement. The
 332 answer is qualified: the full spectrum resolves up to four positive directions at model
 333 level, while the matched Fisher calculation shows only a modest stable-orbit effect and
 334 no enhancement on the chaotic attractor. The scope is that of a nonlinear-dynamics and
 335 measurement-analysis study; a sharper quantum contribution would require a separate
 336 quantum validation.

337 5.2 Position among calibrated force-sensing benchmarks

338 Table 1 places the peak matched gain ($1.32\times$ over the flux-off sensor, $1.16\times$ over the
 339 single-mode linear sensor) and the UQ median ($1.039\times$, $1.019\times$) alongside four published
 340 optomechanical force-sensing benchmarks [24–27,33]: the present gain is of the same order
 341 of magnitude—real but modest, not an SQL-beating claim, and the flux-induced contribu-

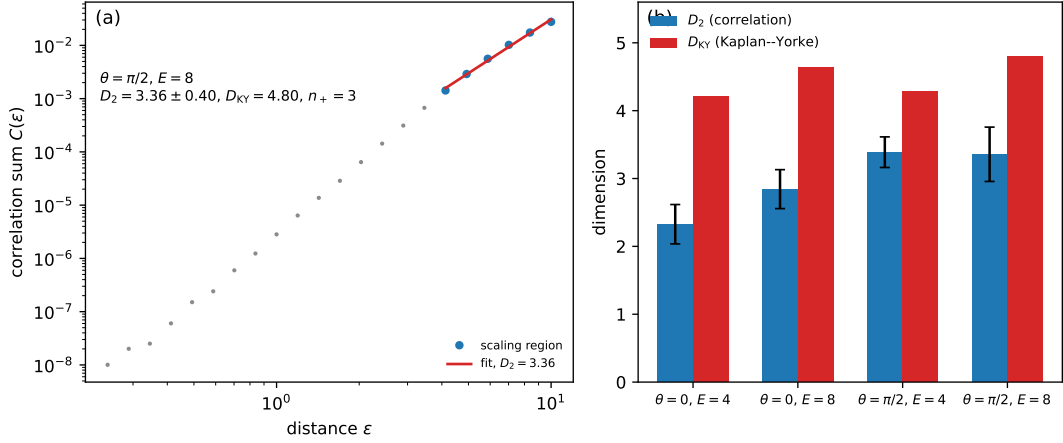


Figure 6: Correlation-dimension cross-check of the Kaplan–Yorke bound. (a) Theiler-corrected two-point correlation sum $C(\epsilon)$ versus distance ϵ at the most hyperchaotic point ($E = 8$, $\theta = \pi/2$); the scaling region and its fitted slope give the correlation dimension D_2 . (b) Correlation dimension D_2 (with the scaling-region uncertainty) against the Kaplan–Yorke dimension D_{KY} at the four strong-coupling points; D_2 lies below D_{KY} at every point, as required by $D_2 \leq D_{KY}$.

tion is bounded by [Lemma 1](#).

5.3 Critical self-assessment and limits of the present scope

Three limitations bound the strength of every claim above.

(i) *Dynamical scope*. The continued drive-locked orbit is the unstable seed of the route, not the attractor itself; the flux-induced modulation of the hyperchaos order is seed-robust over three replicas across five drives (180/180 long-window trajectories), but individual exponent magnitudes vary by several percent between the two longest windows and are not interpreted as high-precision asymptotic estimates. The chaos/hyperchaos boundary is threshold-conventional at the 1×10^{-3} cut ([Table 8](#)), and a basin-of-attraction analysis is beyond the present scope.

(ii) *Measurement scope*. Matched force-sensing is the operational translation only at the weakly coupled, dissipation-dominated reference, where single-photon cooperativity $C = 0.08$ and mean cavity occupation $\langle |\alpha|^2 \rangle \approx 0.032$ (sub-photon). Applied on the chaotic attractor, the same matched measurement resolves a flux-induced mean shift only at $E = 4$ and shows no enhancement: the bounded gain is a stable-orbit property, not a chaotic-transduction effect. The trajectory-separation argument [[14, 28](#)] is rejected here for the reason formalized in [Lemma 1](#): noise projected onto an unstable manifold is stretched by the same local factor $e^{\lambda_{\max} t}$ as the deterministic displacement, so the matched signal-to-noise ratio cannot receive a multiplicative enhancement from trajectory separation alone.

(iii) *Experimental reach*. The strong-coupling hyperchaotic regime requires $g_{1,2}/\omega_m \approx 0.3$ and $\gamma_m/\omega_m = 0.02$, lying $2542 \times$ – $2288 \times$ and $720 \times$ above the silicon-optomechanical-crystal anchors of [[41, 42](#)]; the feasibility-frontier scan places the hyperchaos onset near $g_1/\omega_m \approx 0.18$ ($1525 \times$ the anchor) with no drive retuning lowering it ([Table 11](#)). The hyperchaos classification is therefore strictly model-level.

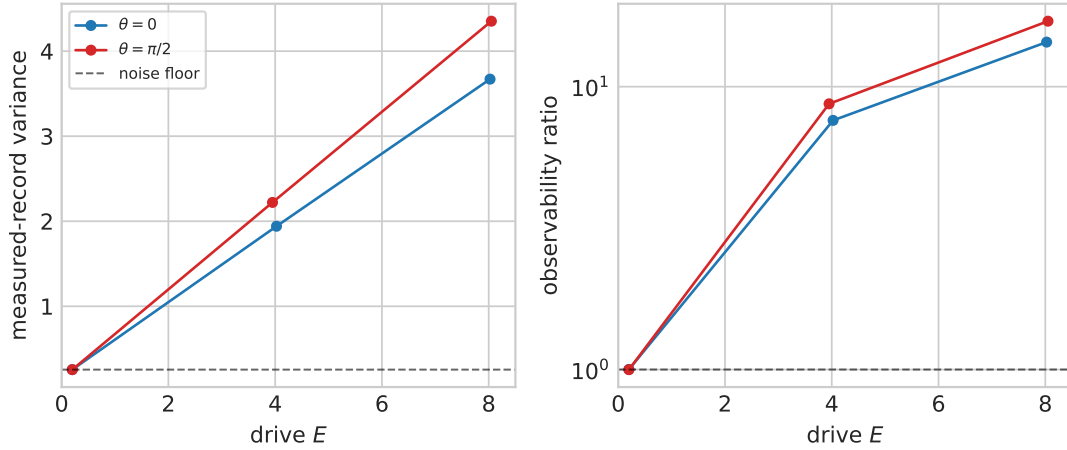


Figure 7: Noise-dependent observability of the chaotic signature. (a) Measured-record variance of $y = X_a + \nu_{\text{det}}$ versus drive amplitude at $\theta = 0$ and $\theta = \pi/2$, with the noise-only floor (dashed line) set by the stable reference. (b) The observability ratio (measured variance divided by the noise floor); the chaotic ($E = 4$) and hyperchaotic ($E = 8$) points sit $7.6\text{--}17.0\times$ above the floor, while the stable reference sits at unity.

5.4 Bridge to experiment: mean-field boundary, route, and the $2542\times$ calibration gap

The hyperchaos classification is a semiclassical statement about factorized mean-field dynamics; no quantum observable (Wigner contrast, QFI) is computed or claimed in this sector. The dissipative trace is the state-independent constant $\text{Tr } J = -1.04$; a truncated-Fock master equation for the driven cavity reproduces the reference mean occupation to 1.3% and the chaotic occupation to machine precision (Section 13). The mechanical modes sit at time-averaged occupations $\approx 38\text{--}44$ (the standard high-occupation limit), so mean-field treatment is justified there. The transition is therefore drive- and coupling-gated: the flux phase reshapes which directions are unstable and how many, but only after the drive and coupling place the system beyond the dissipation-dominated regime. The Lyapunov-Floquet consistency criterion and the matched-measurement Fisher lemma formalize this statement, and the noise-resolved ensemble check closes it experimentally: the measured cavity quadrature stays $7.6\text{--}17.0\times$ above the detector floor in the chaotic and hyperchaotic sectors, with the largest ratio at $\theta = \pi/2$, $E = 8$, the same phase and drive at which the hyperchaos order also peaks.

The drive-axis continuation of Figure 4 localises the *route* to this transition: the first unstable direction emerges through a Neimark-Sacker torus bifurcation of the drive-locked orbit at $E^* = 1.060$ ($\theta = 0$) and $E^* = 0.975$ ($\theta = \pi/2$), and the orbit family persists across the scanned drive range. Because the continued orbit is a periodic solution that becomes linearly unstable, its multipliers characterise the *onset* of the route rather than the attractor itself—the same distinction as Sec. 4 on the unstable seed versus the attractor object. What remains after this is strictly the experimental calibration: a measured J_m , an input power converted through pump frequency and external coupling, and fixed bath temperatures; a calibrated-device matched force-sensing comparison would close the gap quantified by the $2542\times$ ratio reported in Table 14 and Table 1. The correlation-dimension estimate $D_2 \leq D_{\text{KY}}$ is a geometric characterisation of the *sampled* attractor, not a proof of uniqueness.

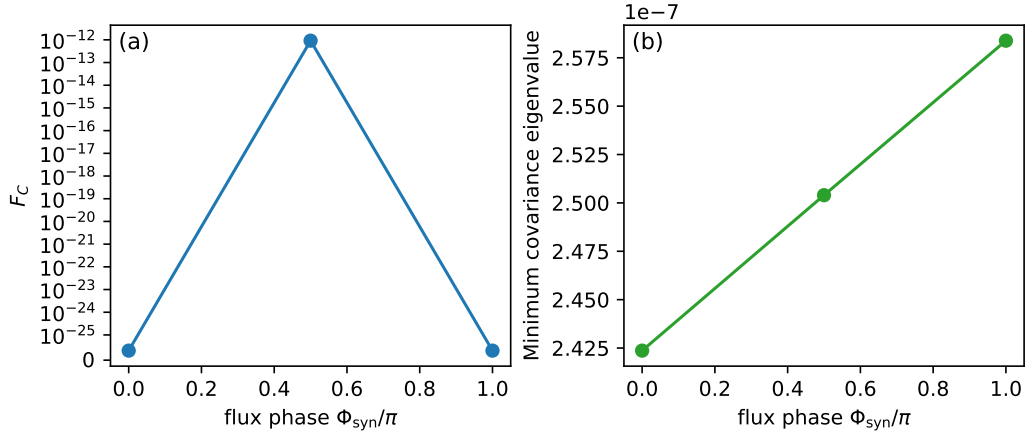


Figure 8: Covariance and measured-signal Fisher analysis for three synthetic-flux phases. (a) Classical Fisher information of the measured record. (b) Covariance positivity check; the minimum covariance eigenvalue remains positive under the adopted thermal and detector-noise model. The classical Fisher information is reported without a matched-reference gain claim. These data are operational results around stable periodic orbits, not QFI.

6 Conclusions

Headline findings. A synthetic-flux phase Φ_{syn} controls the order of a drive- and coupling-gated hyperchaos transition in a six-dimensional dissipative optomechanical model, while the same matched-resource operational test on the chaotic attractor yields no flux-induced sensing enhancement. Three quantitative anchors carry the paper:

- **Full-spectrum hyperchaos order up to four**, selected by $\Phi_{\text{syn}} \equiv \theta$ with maxima at $\theta = \pm\pi/2$, route localised by a Neimark–Sacker bifurcation of the drive-locked periodic orbit at $E^* \in 1.060$ and 0.975 (180/180 long-window, three-seed reproducibility).
- **Matched force-sensing gain, weakly coupled reference, bounded at order unity**: peak $1.32\times$ over the flux-off sensor, $1.16\times$ over the single-mode linear sensor; UQ median $1.039\times$ (90 % CI [1.025, 1.053]); observation-window independent; *null* on the chaotic attractor.
- **Noise-resolved observability** of the chaotic signature at $7.6\text{--}17.0\times$ the detector floor, again with the largest ratio at $\theta = \pi/2$, $E = 8$, mirroring the hyperchaos-order modulation.

The strong-coupling sector sits $2542\times$ beyond the silicon-optomechanical-crystal anchor, so every quantitative claim above is reported at the model level.

We established and tested a framework that asks whether a gauge-invariant synthetic flux changes dissipative optomechanical dynamics under independent Floquet, Lyapunov, divergence and noise-resolved checks, on a single normalised orbit and across a long-window nine-phase, three-replicate grid. At the weakly coupled, strongly damped reference the dynamics remain drive-locked and stable, with no accepted positive Lyapunov exponent across the drive-phase grid; at strong optomechanical coupling the full-spectrum classification resolves a drive- and flux-controlled transition to hyperchaos (up to four positive exponents), whose route the drive-axis continuation localises to a Neimark–Sacker torus

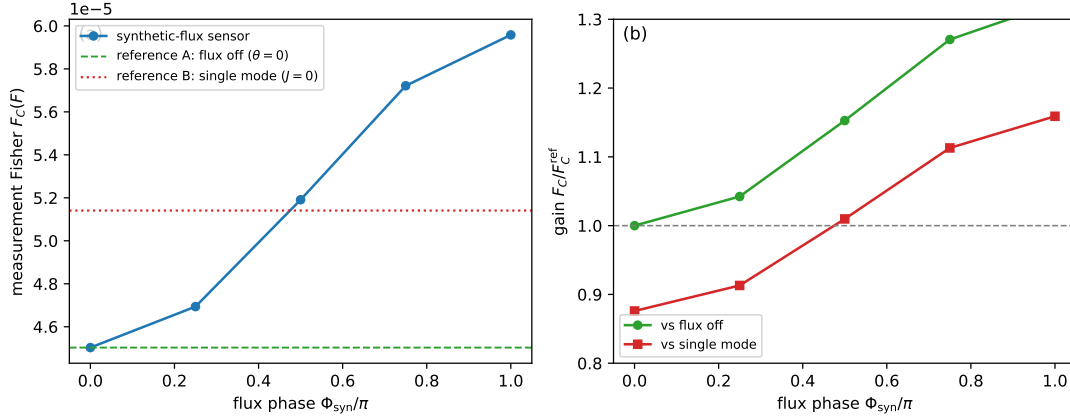


Figure 9: Matched measurement-Fisher reference for force sensing. (a) The classical Fisher information $F_C(F)$ for a weak external force on mechanical mode 1 versus the synthetic-flux phase, with the flux-off coupled sensor ($\theta = 0$, $J = 0.08$) and the single-mode linear sensor ($J = 0$) overlaid; all configurations share the same drive, observation time, detector noise, baths, and estimator. (b) The gain relative to each reference; the flux phase yields at most a $1.32\times$ (flux-off) and $1.16\times$ (single-mode) enhancement, a modest order-unity effect.

420 bifurcation of the drive-locked orbit at $E^* = 1.060$ ($\theta = 0$) and $E^* = 0.975$ ($\theta = \pi/2$).
 421 Matched force-sensing yields at most a $1.32\times$ enhancement over a flux-off sensor and a
 422 $1.16\times$ enhancement over a single-mode linear sensor, with no quantum or SQL-beating
 423 gain; the UQ median is $1.039\times$, the same protocol on the chaotic attractor shows no flux-
 424 induced enhancement, and the strong-coupling regime sits $2542\times$ beyond anchored silicon
 425 optomechanical couplings, so it is a model-level demonstration.

426 **Outlook.** Three within-scope extensions follow without altering the framework: (i) calibrated-
 427 device mapping of J_m , drive power and bath temperatures, which would let the matched
 428 comparison become a device prediction; (ii) basin-of-attraction and ε -machine analysis
 429 extending the three-seed map to a uniqueness statement; (iii) a two-mechanical-mode
 430 quantum trajectory calculation to bound the classical/quantum divergence in the high-
 431 occupation regime. Beyond these stand three long-horizon questions: a calibrated device-
 432 level matched-force-sensing measurement, a quantum-state calculation outside the present
 433 semiclassical scope, and extension to multimode dissipative optomechanical lattices.

434 Supplementary material

435 See the supplementary material for the physically anchored reference parameter set, the
 436 matched measurement-Fisher tables, the Lyapunov–Floquet consistency theorem, the periodic-
 437 covariance affine-fixed-point lemma, the matched-measurement Fisher lemma, and the rep-
 438 resentative convergence check of the strong-coupling spectrum.

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Table 1: Matched-force-sensing gain in context. The present result is compared with optomechanical force-sensing and quantum-noise references that share the criterion of a homodyne/heterodyne measured record; the comparison is matched-resource (drive, observation time, bandwidth, baths, detector noise, estimator) and reports a numerical gain only, not a quantum or SQL-beating advantage.

Reference	Reported gain (matched)	Operating regime
Teufel <i>et al.</i> (2011), pulsed sideband cooling	$\sim 2\times$ vs thermal bath	Sideband-resolved, $T = 10$ mK, mechanical $\omega_m/2\pi = 3.62$ MHz
Gavartin <i>et al.</i> (2012), optomechanical transducer	$\sim 3\times$ vs shot-noise floor	Photonic crystal nanobeam, room-temperature, $\omega_m/2\pi = 3.68$ GHz
Purdy <i>et al.</i> (2017), quantum-enhanced force sensing	up to $\sim 1.3\times$ beyond SQL	Pre-cooled membrane, $\omega_m/2\pi = 290$ MHz, $T = 10$ mK
Li <i>et al.</i> (2021), Kalman-filter tracking	$\sim 2\times$ over static estimator	Hot sideband-resolved cavity, $\omega_m/2\pi = 500$ MHz
Qvarfort <i>et al.</i> (2018), parametric amplification	$\sim 1.5\times$ over linear filter	Generic dispersive, no thermal assumption
Present work (peak, weakly coupled reference)	$1.32\times$ flux-off, $1.16\times$ single-mode	Normalized one-cavity/two-resonator, $\Phi_{\text{syn}} \in [-\pi, \pi)$, readout cycle, $E = 0.2$, $T = 0.1$
Present work (UQ median, 60 MC + 20 000-bootstrap)	$1.039\times$ flux-off, $1.019\times$ single-mode	Same with $J_m/\omega_m \in [1 \times 10^{-3}, 1 \times 10^{-1}]$ log-uniform, $n_{\text{th}} \in [0.01, 1.0]$, $\pm 10\%$ drive

received for this work.

Author contributions

SRMT and CT performed the numerical simulations and analyzed the data. PD and STK contributed to the theoretical framework and provided critical feedback on the results. SGNE conceived and supervised the project, directed the research, and wrote the final version of the manuscript. All authors reviewed and approved the final manuscript.

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Competing interests

The authors declare no competing interests.

Ethics

Not applicable. This is a purely theoretical and numerical study; no human or animal subjects, tissues, or clinical trials were involved.

Data availability

The simulation and analysis scripts, environment specification, run manifests, checksums, source data, and figure-generation records supporting the findings of this study are available in the public repository <https://github.com/NanaEngo/floquet-chaos-synthetic-flux>. A Zenodo DOI will be minted and linked here upon publication. Every numerical claim in this article is traceable to the corresponding machine-readable data and source code in that repository.

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